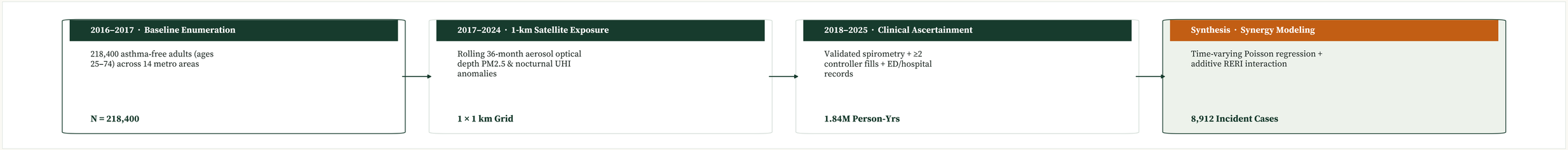
POPULATION SURVEILLANCE COHORT · 1.84M PERSON-YEARS · STROBE · SYNTHETIC DATA

Long-Term PM2.5 Exposure, Urban Heat Islands, and Incident Adult-Onset Asthma Across 14 Metropolitan Cohorts (2016–2025)

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01 Public Health Context & Aim

Fine particulate pollution (PM2.5) and urban heat island (UHI) warming co-concentrate in dense metropolitan neighborhoods, amplifying airway oxidative stress [1].

• **Objective:** Quantify independent and joint associations of chronic PM2.5 and nocturnal UHI intensity with adult-onset asthma across 1.84 million person-years.

02 Exposure & Statistical Methods

• **Exposures:** Ensemble satellite-monitor PM2.5 (μg/m^3) and warm-season nocturnal surface temperature anomaly (°C vs. rural buffer).

• **Models:** Multivariable quasi-Poisson survival models adjusted for age, sex, smoking pack-years, BMI, greenness (NDVI), and neighborhood deprivation index.

• **Effect Modification:** Relative Excess Risk due to Interaction (RERI) quantified departure from additivity [2].

Synergistic Air-Heat Respiratory Risk

IRR 1.38 in High-Heat Quartile (RERI = 0.21; p = 0.003)

Overall incidence was 4.84 per 1,000 person-years (8,912 cases). Each +10 μg/m^3 PM2.5 increased asthma risk by 24% overall (IRR 1.24, 95% CI 1.18–1.31), rising to 38% in high-UHI neighborhoods.

03 Exposure–Response Incidence by Heat Stratum

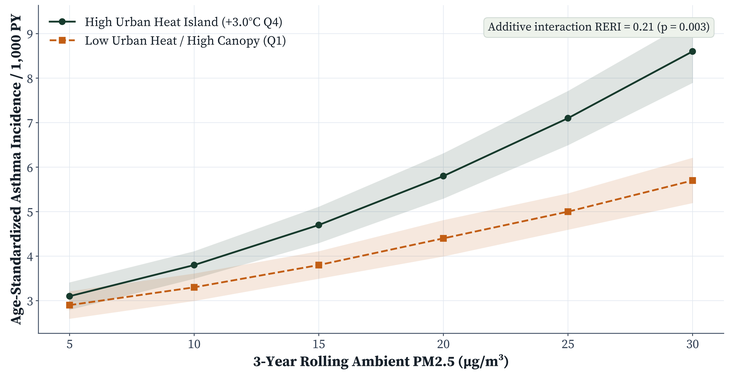


Figure 1. Standardized adult-onset asthma incidence per 1,000 person-years across chronic PM2.5 and nocturnal UHI quartiles.

04 Stratified Multivariable Models

Exposure Stratum	Cases / PY	Adjusted IRR (95% CI)	p-value
Overall (+10 μg/m³ PM2.5)	8,912 / 1.84M	1.24 (1.18–1.31)	0.0001
Low UHI Q1 (<+0.8°C)	1,640 / 0.46M	1.14 (1.07–1.22)	0.0004
Moderate UHI Q2–Q3	4,192 / 0.92M	1.23 (1.16–1.30)	0.0001
High UHI Q4 (≥+3.0°C)	3,080 / 0.46M	1.38 (1.29–1.48)	0.0001
Never-smokers sensitivity	5,410 / 1.19M	1.26 (1.19–1.34)	0.0001

Adjusted for age, sex, smoking, BMI, NDVI greenness, and area deprivation.

05 Limitations & Policy Implications

• **Limitation:** Residential geocoding omits workplace indoor air filtration and transit exposure.

• **Policy Action:** Co-targeting PM2.5 emissions and urban heat mitigation in high-UHI districts could avert an estimated 19.4% of attributable adult-onset cases.

References: [1] Liu C et al. N Engl J Med 2019;381:705–715. [2] VanderWeele TJ, Knol MJ. Epidemiol Rev 2014;36:115–147.

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